



Food and Agriculture Organization
of the United Nations

PyAEZ v3.0

**Dario Spiller, Shahla Asgharinia, Rutendo Mukaratirwa,
Federica Chiozza, and Livia Peiser**

Geospatial Unit, Land and Water Division (NSL)



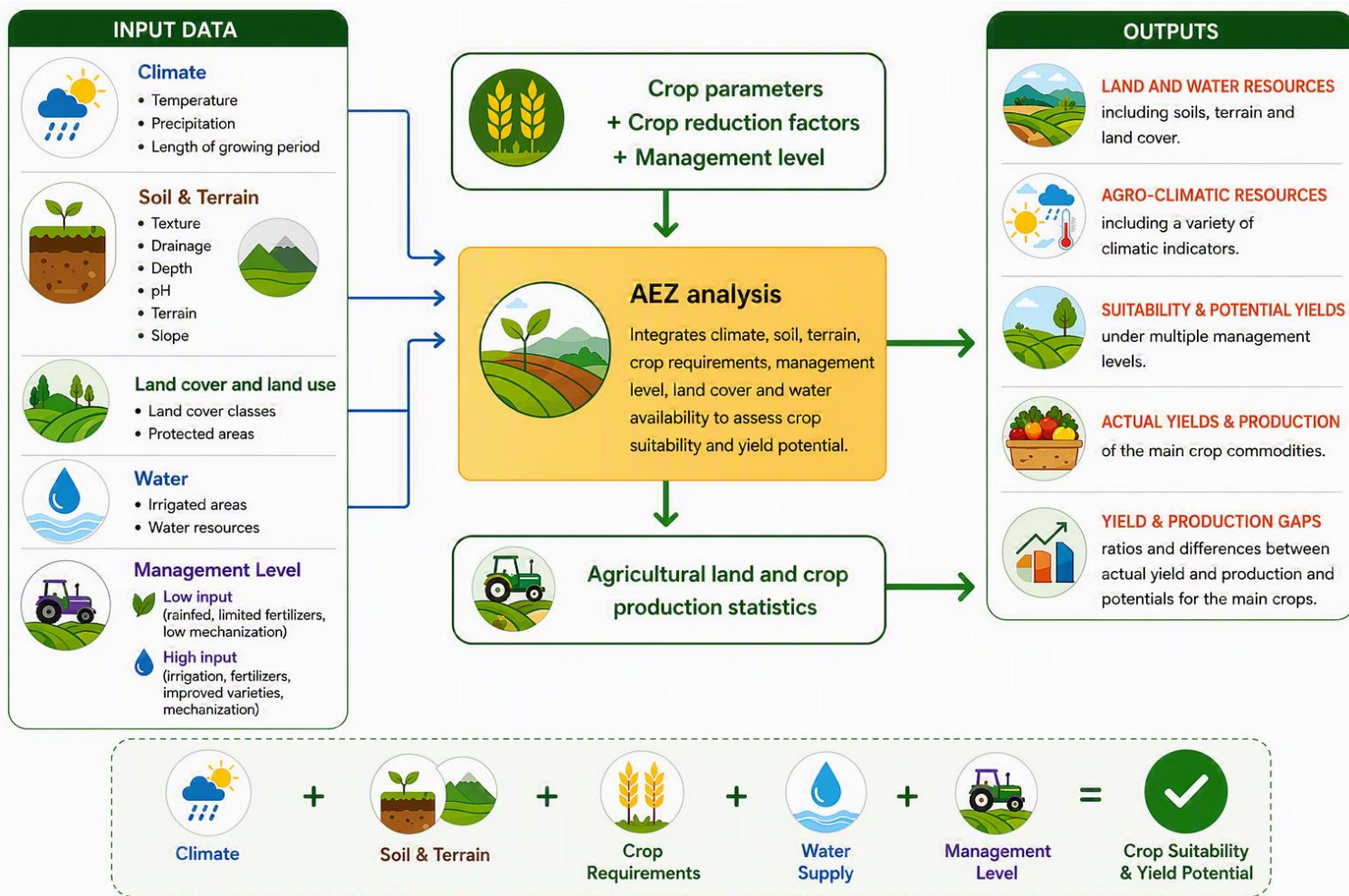
PyAEZ is an open and modular Python implementation of the Agro-Ecological Zoning (AEZ) framework. It provides transparent, code-based workflows for reproducing, testing, adapting and extending AEZ assessments. PyAEZ is designed to make AEZ calculations more accessible, reproducible and adaptable, particularly for national and subnational applications.

- **PyAEZ** development was started in **2022** by the **Asian Institute of Technology (AIT)** in collaboration with **FAO-RAP**, the **FAO NSL Geospatial Unit**, and the **SAMIS Project** to support national Agro-Ecological Zone (AEZ) implementation in Lao PDR.
- **2022–2024:** Development continued through collaboration with **AIT** and **IIASA**, resulting in **PyAEZ v2.x**.
- **From 2024:** **Mississippi State University (MSU)** contributed to code optimization and performance improvements, significantly reducing computational time.
- **2025–present:** Development and maintenance of **PyAEZ** are led internally by the **FAO NSL Geospatial Unit**. **PyAEZ v3.0** aligned with the updated **GAEZ v5** methodological framework **migrated to the new GitHub repository**
<https://github.com/un-fao/pyaez>



The Agro-ecological Zoning Framework

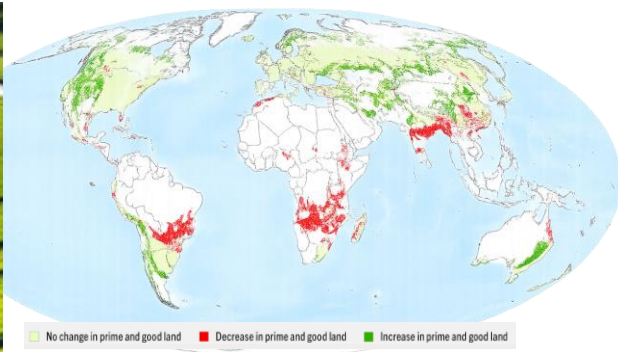
The Agro-Ecological Zones (AEZ) methodology is a framework based on the principles of land evaluation (FAO, 1976) to support sustainable agricultural development by characterizing agricultural land utilization options





Agro-ecological Zoning Applications

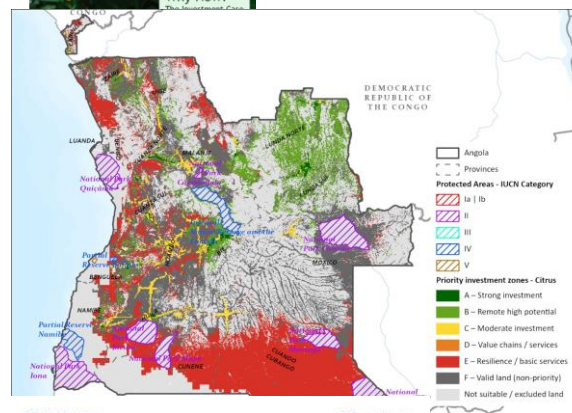
Global applications



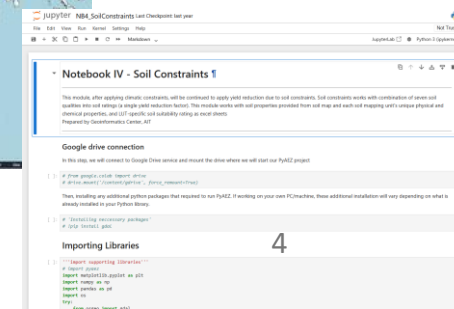
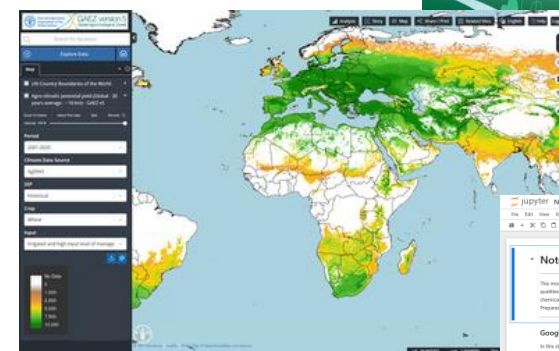
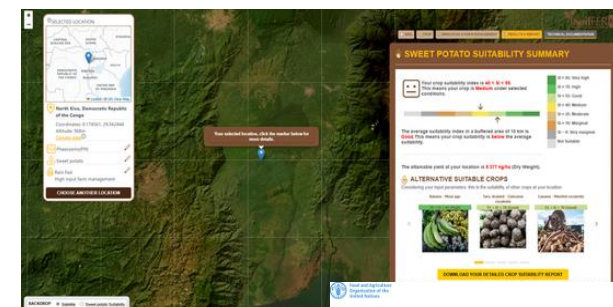
National applications



Spatial analysis and investment targeting



Tools





Global Agro-Ecological Zoning v5 – GAEZ

GAEZ is an application at global scale, updated recently to version 5. All data is freely accessible on the GAEZ v5 platform

Notebook 0 — Input Data Preparation

PyAEZ Version 3.0 | 2025

This notebook guides users through the preparation and organization of all input datasets required for running PyAEZ.

Prepared by: *Stefano Iglietta*

Geospatial Unit — Land and Water Division (WLD)

Food and Agriculture Organization of the United Nations (FAO)

Python Agro-Ecological Zoning – PyAEZ

PyAEZ allows user to run local agro-ecological zoning analysis using user-defined input data and user-defined spatial resolution.

Input data from the user: Area of Interest and scenario

When should I use GAEZ vs PyAEZ?

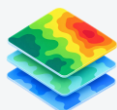
GAEZ and PyAEZ serve complementary purposes. The choice is based on purpose:

Use GAEZ when you need...



Global reference information

Ready-to-use harmonized global datasets and outputs



Standard GAEZ results

Apply the established GAEZ methodology and crop requirements



Quick access to existing products

Explore existing suitability and AEZ products without rebuilding the workflow



Cross-country consistency

Use a common global reference for comparison

Use PyAEZ when you need...



National or local analysis

Use national/local climate, soil, terrain or other datasets



A customizable AEZ workflow

Modify crop parameters, requirements, inputs or individual processing steps



Testing, validation and development

Inspect intermediate results, test assumptions and compare against GAEZ



Scenario and case-study flexibility

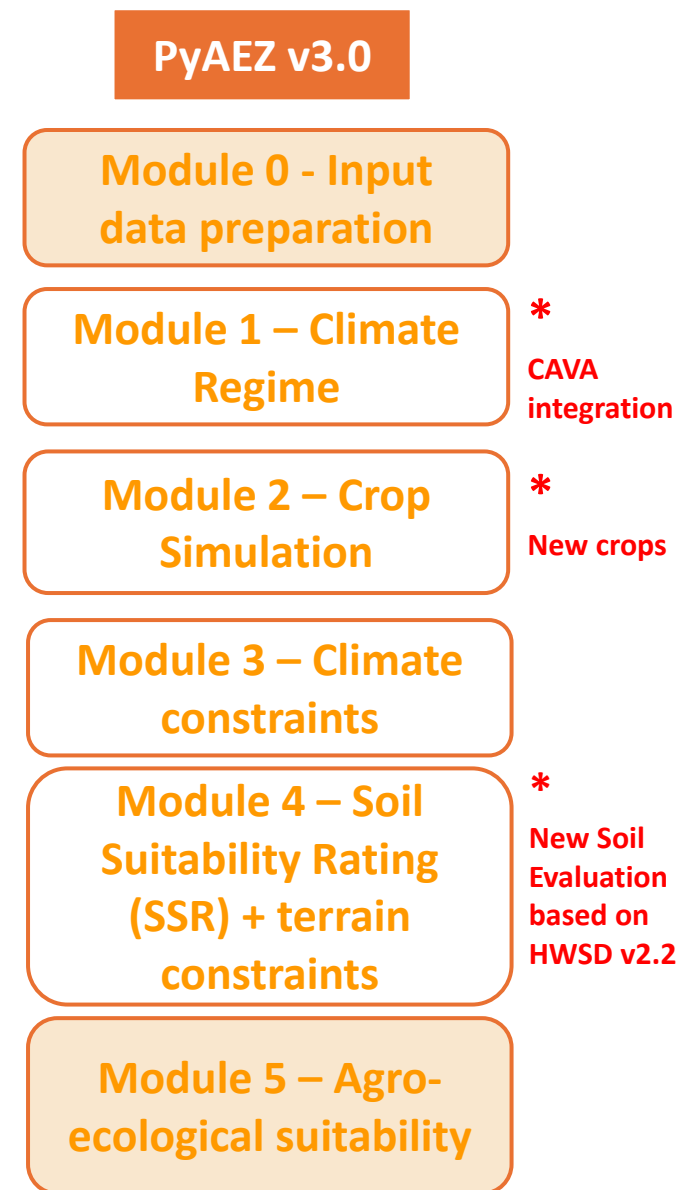
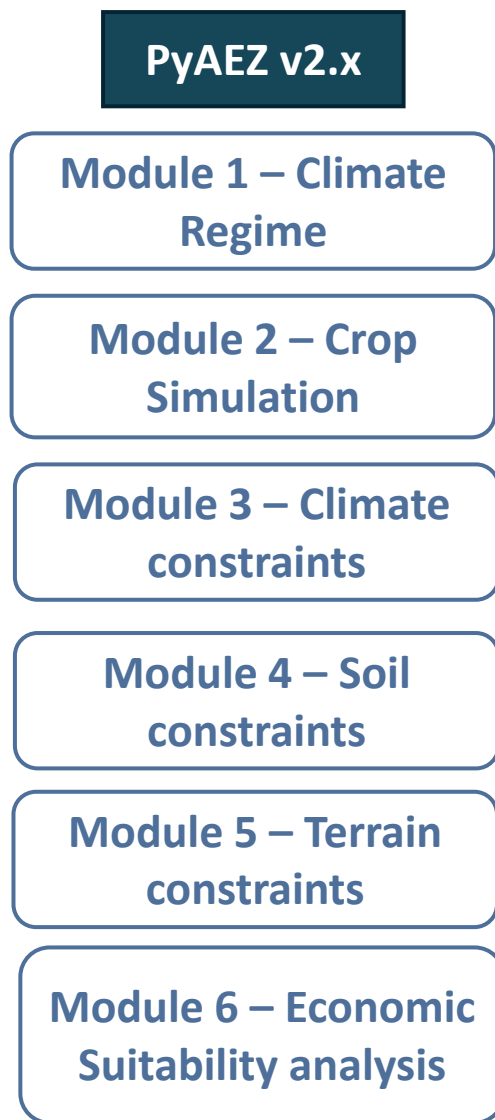
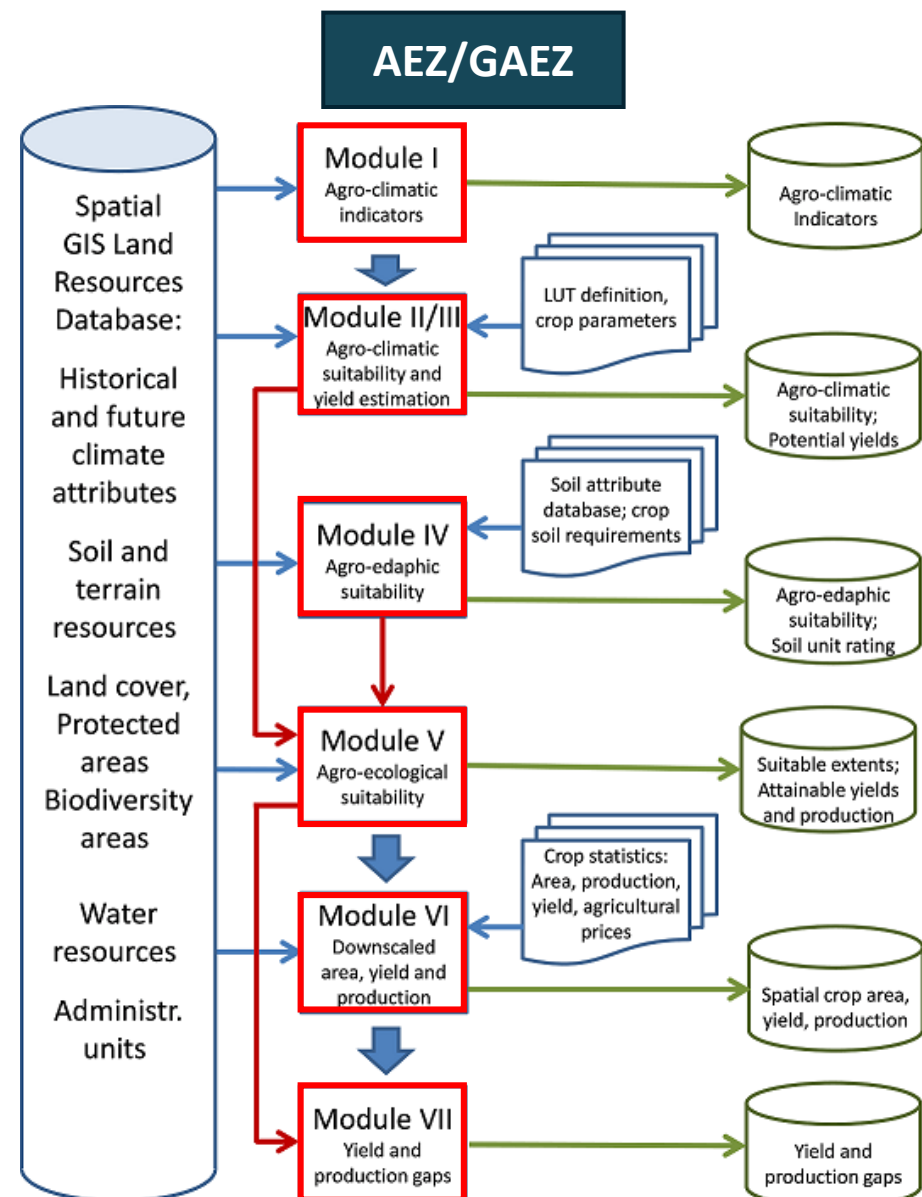
Adapt the workflow to specific countries, crops, datasets or scenarios

GAEZ = the global reference PyAEZ = the adaptable implementation.

Use PyAEZ when the question requires going beyond standard global inputs or outputs

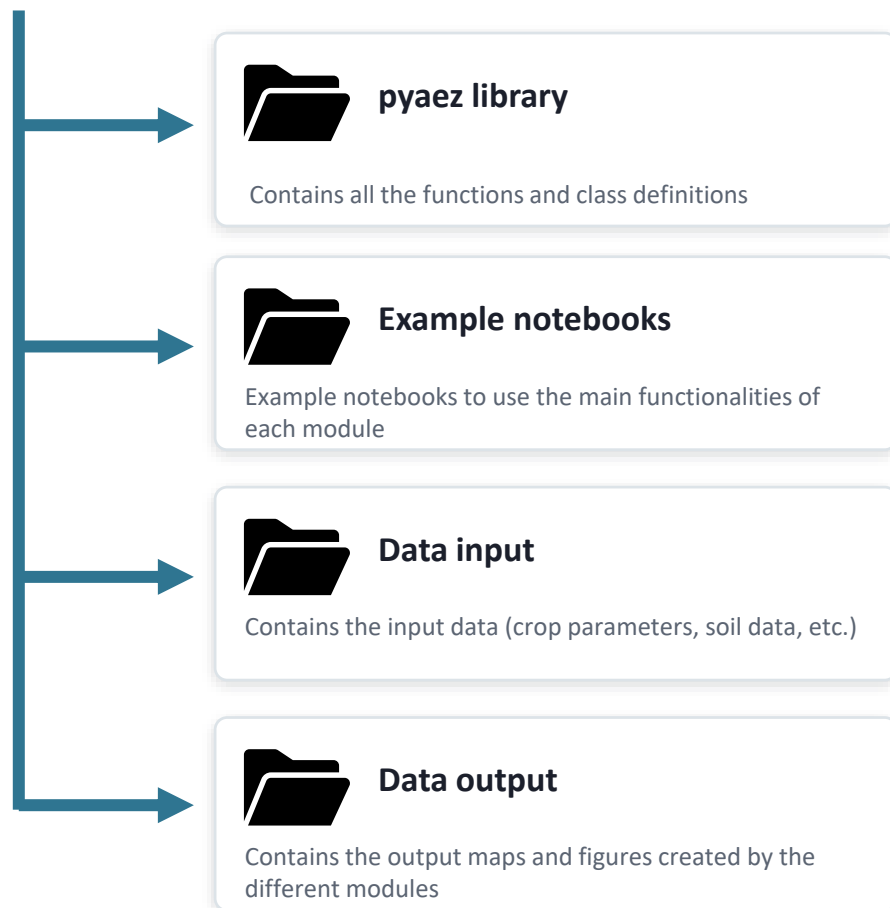


Modular approach



The PyAEZ software is organized in the following main folders

PyAEZ 3.0



PyAEZ is written in Python. The code is made available on GitHub and could be used at different levels.

- ***Expert users*** may work directly with the python code using/modifying the example notebooks and referring to the technical documentation where the functions and classes are described.
- ***Users without expertise in python*** might still refer to the notebooks.
- Next steps will explore the possibility to run the notebook on a dedicated server with pre-installed environment or develop a Graphical User Interface (GUI) facilitating the user experience.



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	reviewed example data input	4c86ce6 · 3 hours ago	200 Commits
PyAEZ3.0	reviewed example data input	3 hours ago	
.gitattributes	solved minor issues	3 hours ago	
.gitignore	Update package configuration	6 hours ago	
CITATION.cff	Revise CITATION.cff with new author and repo details	yesterday	
MANIFEST.in	moving init file and toml file	4 hours ago	
README.md	Revise contributions and add useful links section	yesterday	
environment_pyaез.yml	Update environment_pyaез.yml	last month	
pyproject.toml	updated toml file, moved init file	4 hours ago	

README

PyAEZ

PyAEZ open-source Python 3.x Developed by FAO

PyAEZ is an open and modular Python implementation of the Agro-Ecological Zoning (AEZ) framework.

It provides transparent, code-based workflows for reproducing, testing, adapting and extending AEZ assessments. PyAEZ is designed to make AEZ calculations more accessible, reproducible and adaptable, particularly for national and subnational applications.

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Languages





PyAEZ v3.0

**Module 0 - Input
data preparation**

**Module 1 – Climate
Regime**

**Module 2 – Crop
Simulation**

**Module 3 – Climate
constraints**

**Module 4 – Soil
Suitability Rating
(SSR) + terrain
constraints**

**Module 5 – Agro-
ecological suitability**

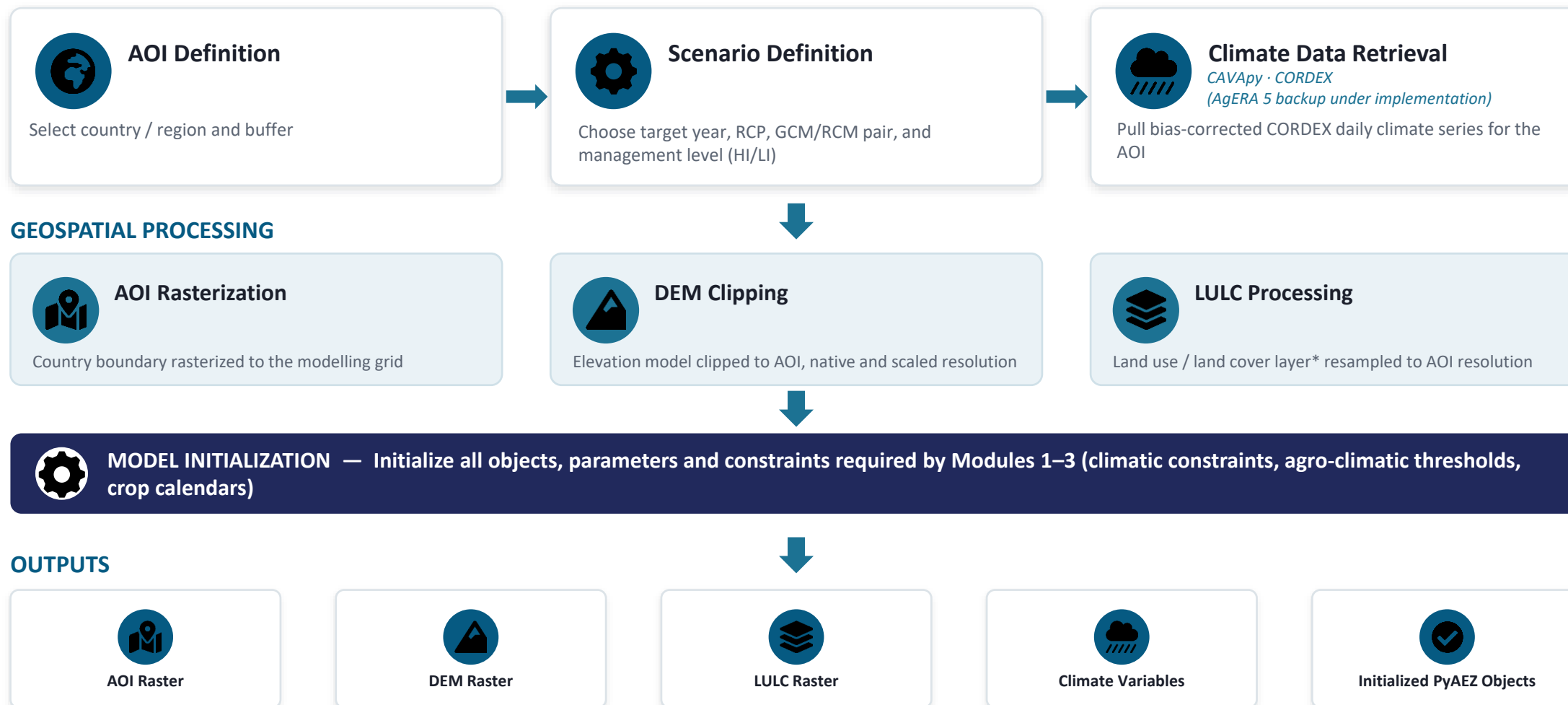
The entry point for PyAEZ users is Module 0.

Here, the users can:

- *Choose the country or region for the analysis*
- *Retrieve automatically the climate data*
- *Choose the crop from a pre-defined list of crops (based on GAEZ v5)*
- *Modify the parameters or enter user-defined input data*

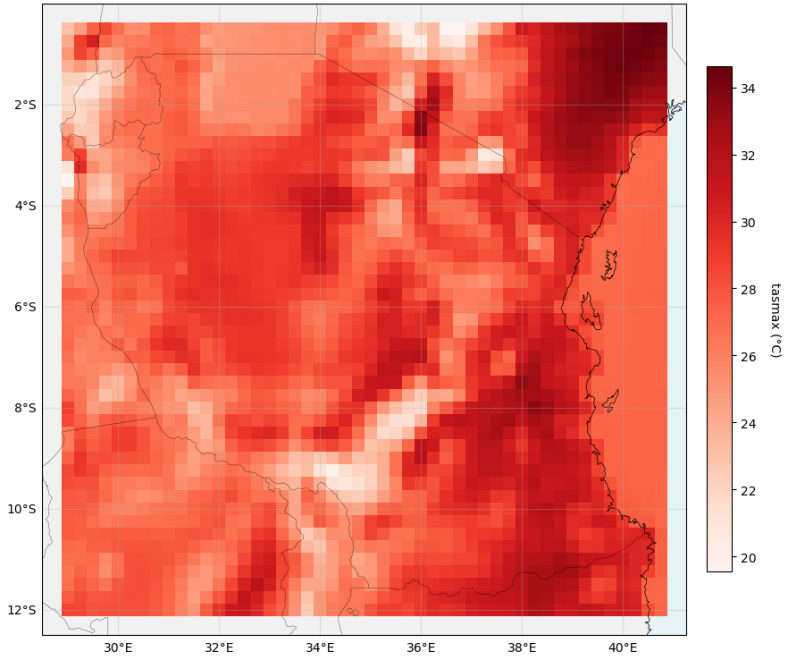
Based on the computation made in Module 0, the other modules do not require any additional input from the users (as shown in the example notebooks).

Prepare, harmonize and validate every input required by the PyAEZ workflow before running the specific modules.

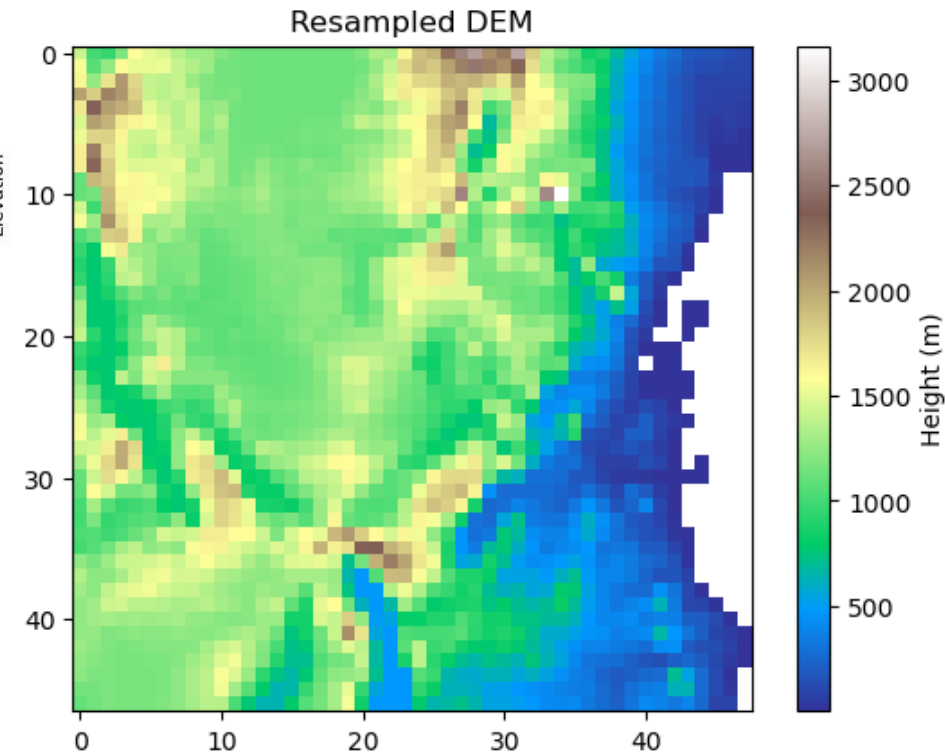
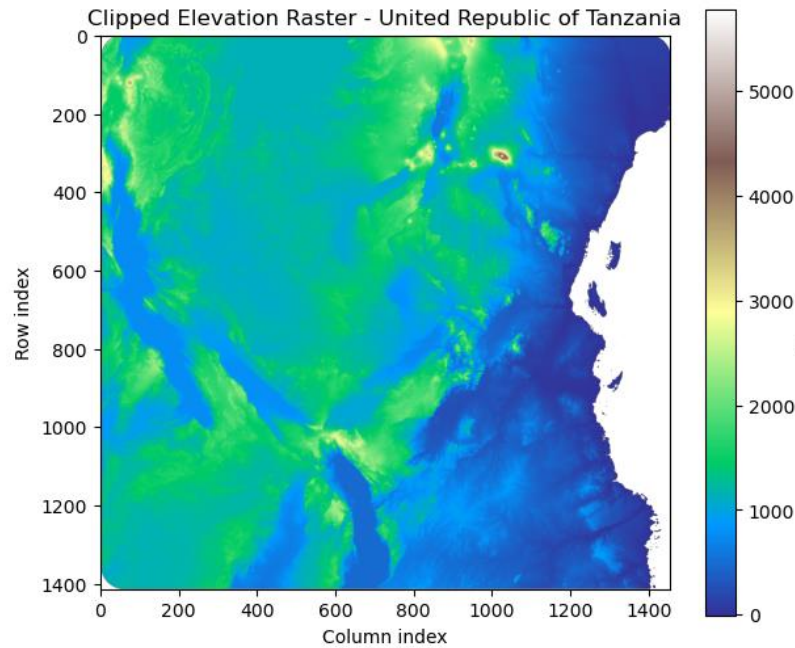


* The dataset soil_regime_CRUTS32_Hist_8110, available under the **Agro-ecological Zones** sub-theme of **GAEZ v4**

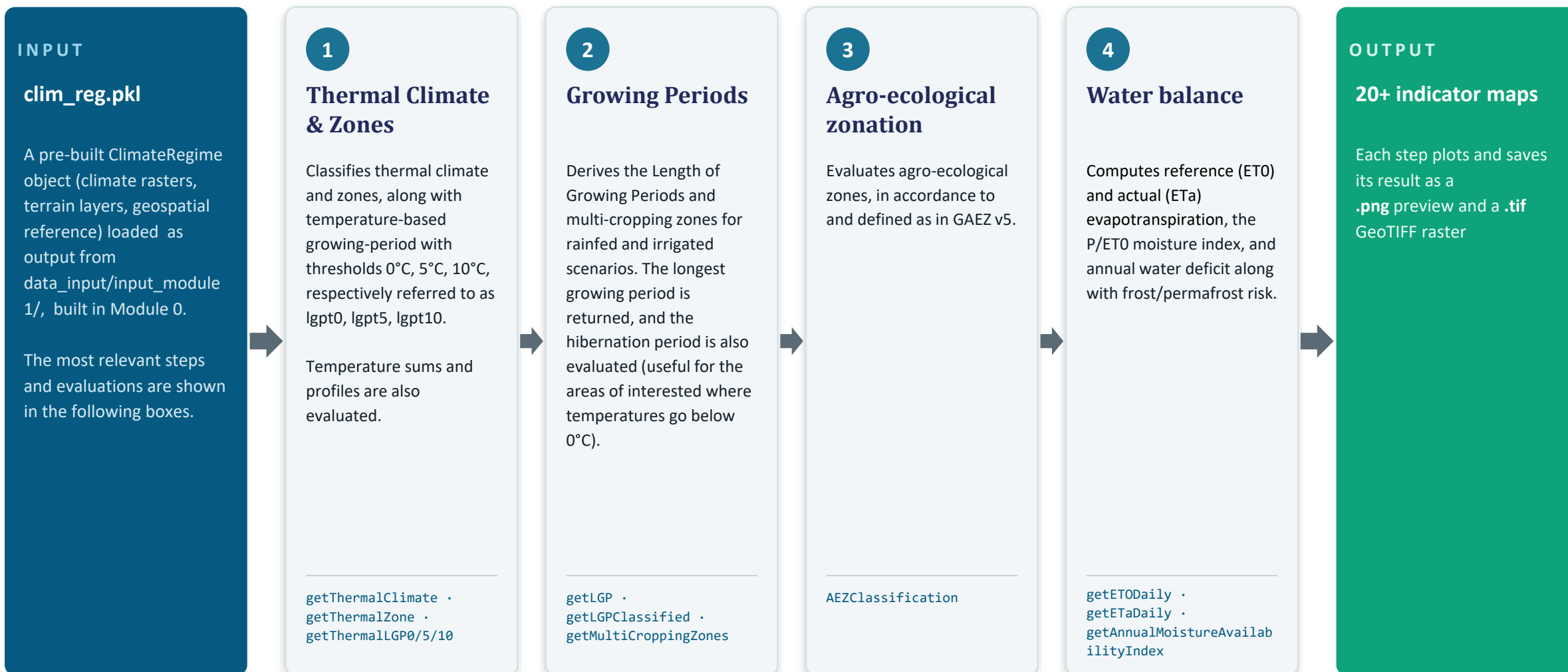
Mean Max Temperature

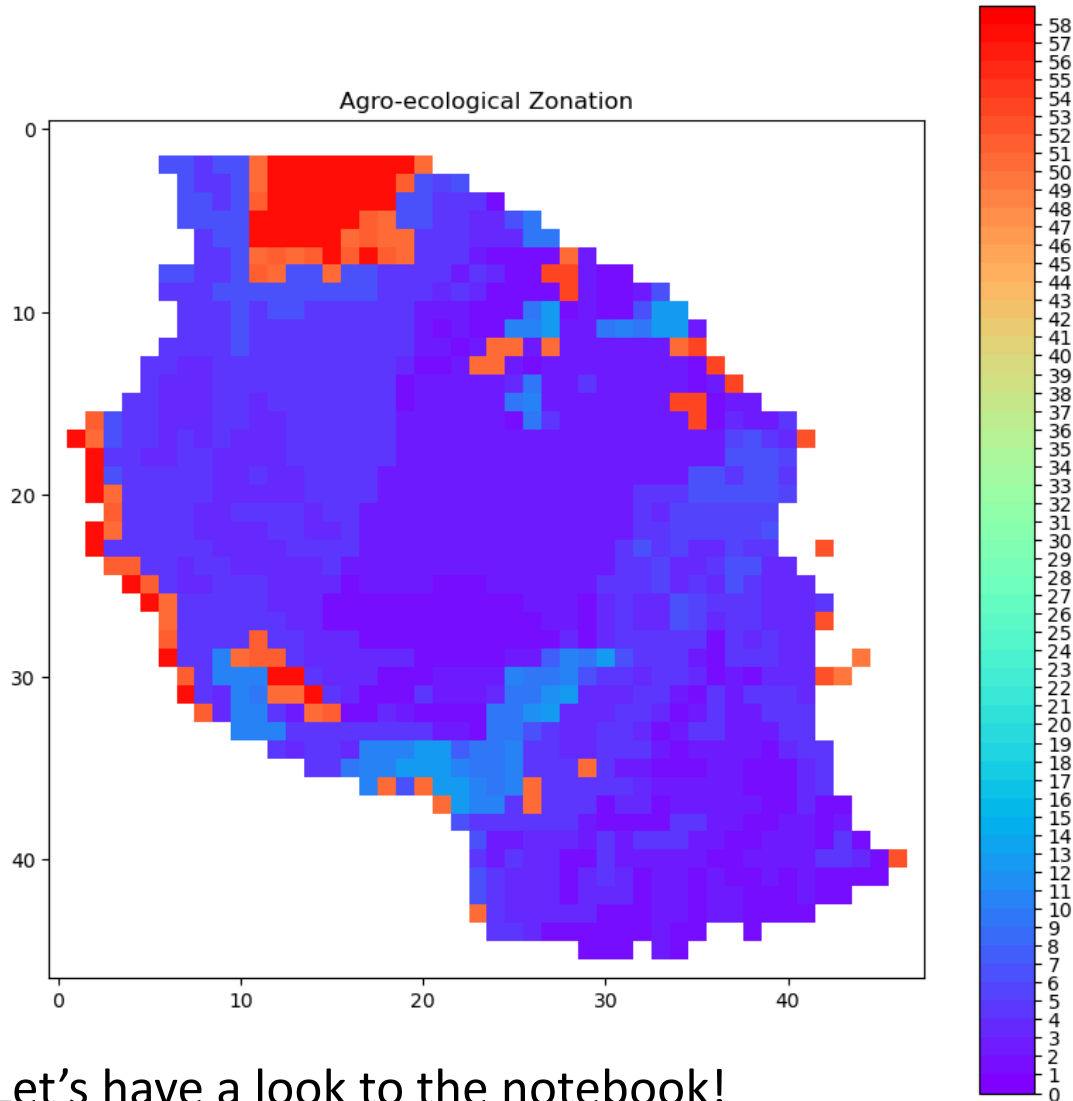


Example outputs from module 0...

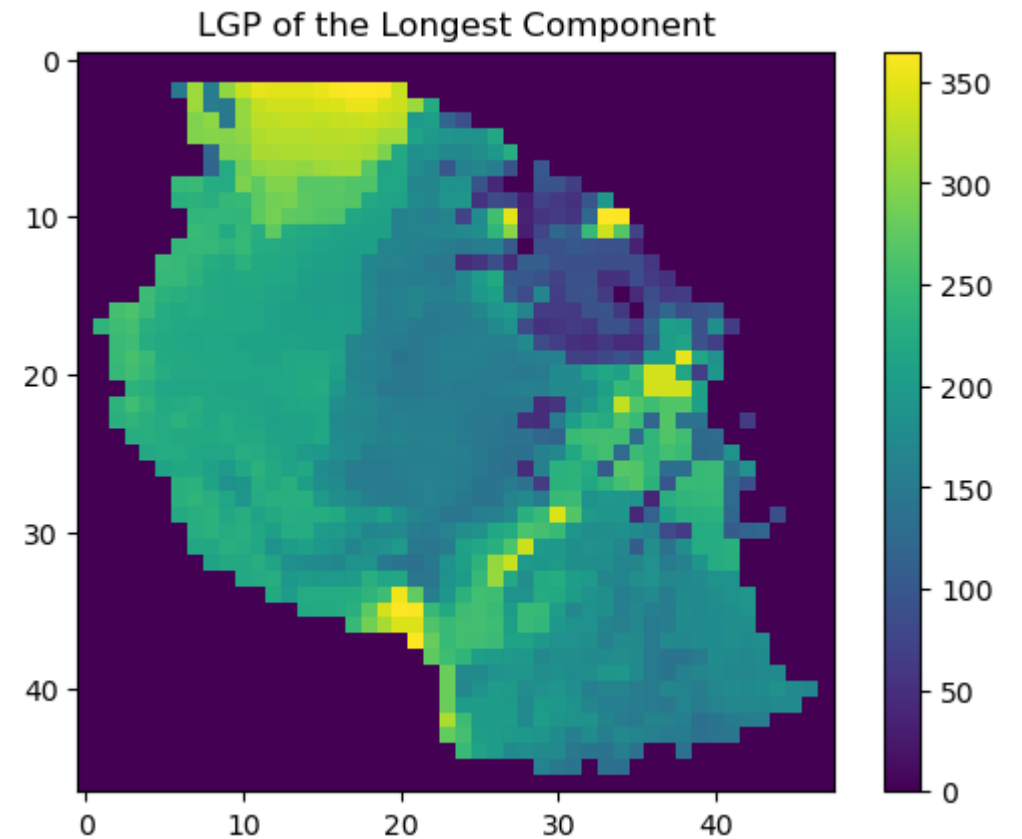


Let's have a look to the notebook!

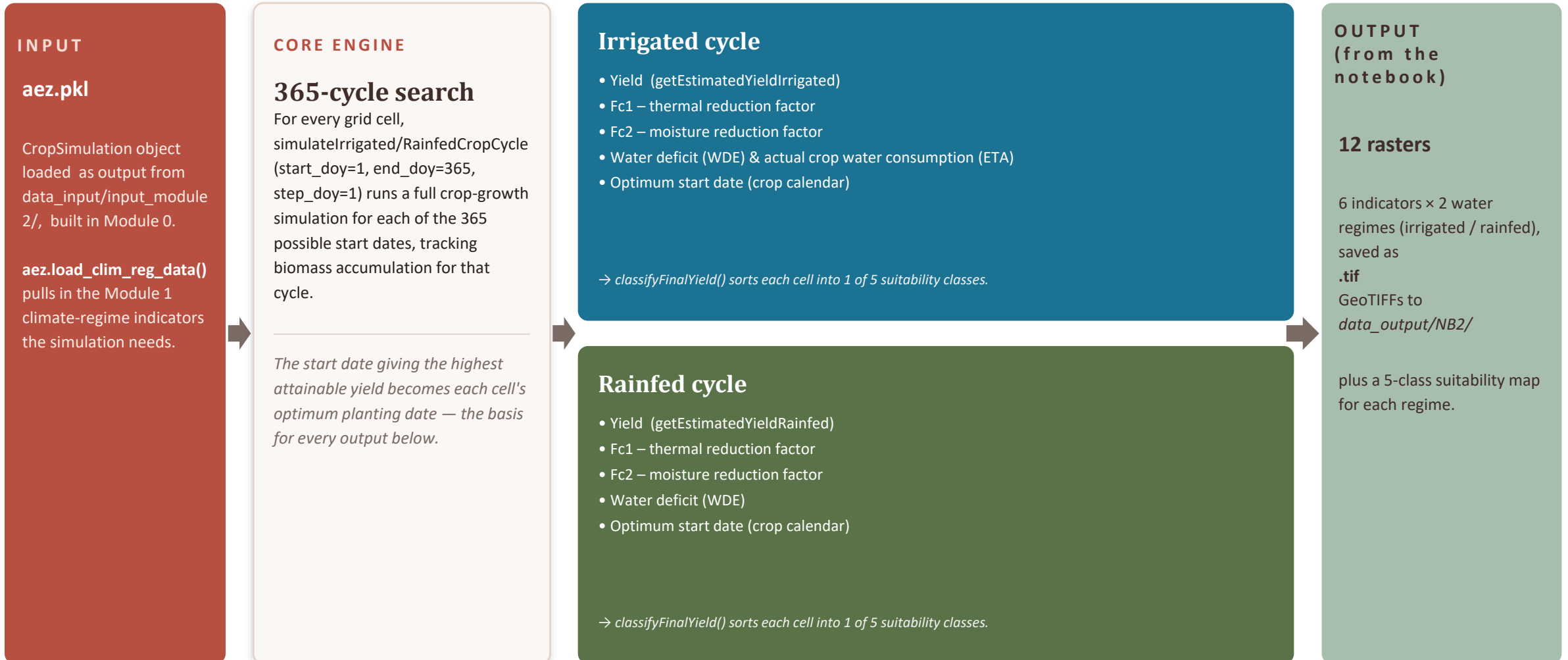




Example outputs from module 1...



Let's have a look to the notebook!



INPUT `clim_con.pkl` — ClimaticConstraints object from
data_input/input_module3/, built in Module 0



CORE ENGINE

`clim_con.compute_yield(condition_type)` For each grid cell, retrieves the agroclimatic length-of-growing-period class (`lgp_agc`) and looks up the matching crop-specific reduction table from the GAEZ appendices to derive `fc3` — the climatic constraint factor. Multiplying `fc3` by Module 2's maximum attainable yield gives the climate-adjusted yield for that condition.



Irrigated condition

```
clim_yield_irr, fc3_irr = clim_con.compute_yield('irrigated')
```

- Climate-adjusted yield (`clim_yield_irr`)
- `fc3_irr` — climatic reduction factor for the irrigated cycle



Rainfed condition

```
clim_yield_rain, fc3_rain = clim_con.compute_yield('rainfed')
```

- Climate-adjusted yield (`clim_yield_rain`)
- `fc3_rain` — climatic reduction factor for the rainfed cycle

Saved to data_output/NB3/: `{crop}_yld_rain_m3.tif · fc3_{crop}_rain.tif · {crop}_yld_irr_m3.tif · fc3_{crop}_irr.tif · lgpagc.tif`

fc3 — fraction (0–1) of the maximum attainable yield retained once climatic constraints are applied; derived per agroclimatic LGP class (`lgp_agc`) from the GAEZ reduction-factor appendices. Optional: re-run for a named crop via `*CropSpecific()` variants.



PyAEZ v3.0 · Module 4

Soil Quality Ratings (SQ1–SQ7) → SSR + Terrain Constraints



Example output

Module 0



Module 4

Country

AOI

Tanzania

Crop

90 selectable crops

Maize

Water supply

rainfed / sprinkler / drip / gravity

Rainfed

Farming

conventional / organic

Conv

Input

low / intermediate / high

High





SQ1–SQ7: what each rating represents?

Each soil quality is crop-specific and expressed on the 0–100 AEZ suitability scale.

SQ1

Nutrient availability

SQ2

Nutrient retention

SQ3

Rooting conditions

SQ4

Oxygen availability

SQ5

Salinity / sodicity

SQ6

Calcium carbonate and gypsum

SQ7

Workability

SQ1–SQ7: what each rating represents

Each soil quality is crop-specific and expressed on the 0–100 AEZ suitability scale.

SQ1	Nutrient availability	texture · OC · pH · TEB
SQ2	Nutrient retention	texture · BS · CEC
SQ3	Rooting conditions	root depth · texture · coarse fragments · phases
SQ4	Oxygen availability	drainage · phases · impermeable layers
SQ5	Salinity / sodicity	EC · ESP · phases
SQ6	Calcium carbonate and gypsum	CaCO ₃ / gypsum context · phases
SQ7	Workability	root depth · fragments · drainage · phase · texture

Rating principle

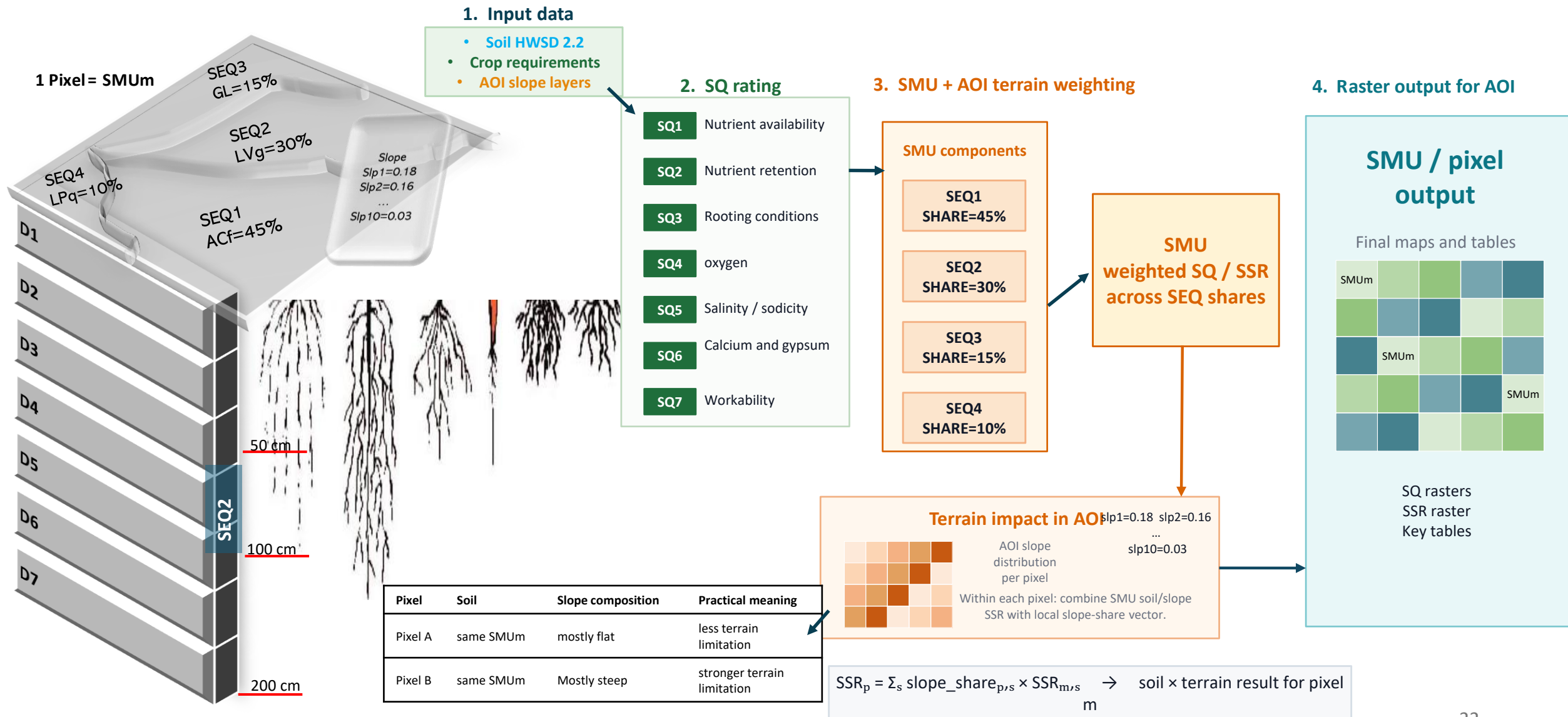
Every attribute is first rated:
 $\tau(\text{attribute}) \rightarrow 0 \dots 100$

Layer ratings are combined with FSQ.
Soil-unit ratings are then aggregated
using component shares and layer depth rules.

AWC

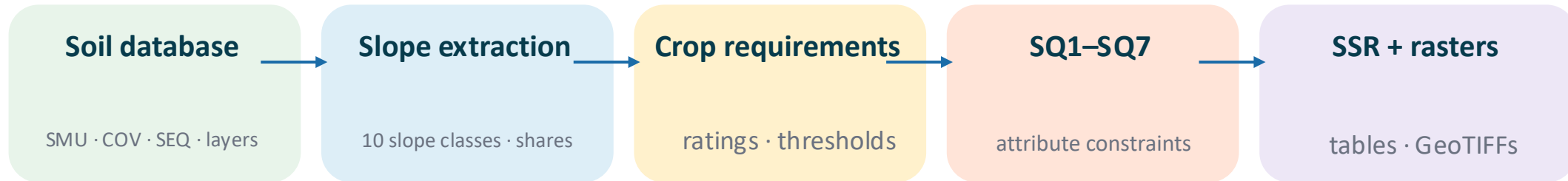
Available water holding capacity

Ref_AWC. Texture. root depth · fragments · phase



What Module 4 is doing?

Convert soil and terrain information into crop-specific soil suitability ratings on a 0–100 scale.



Step 1:

Extracts soil–slope data

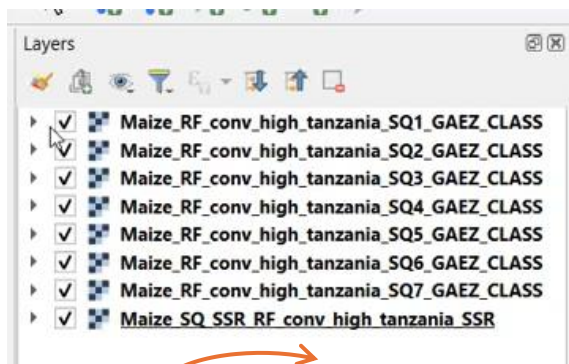
Step 2:

Rates soil qualities SQ1–SQ7 for each crop, water regime, input level and soil component.

Final **SSR** uses both component composition and pixel-level local slope-share rasters.

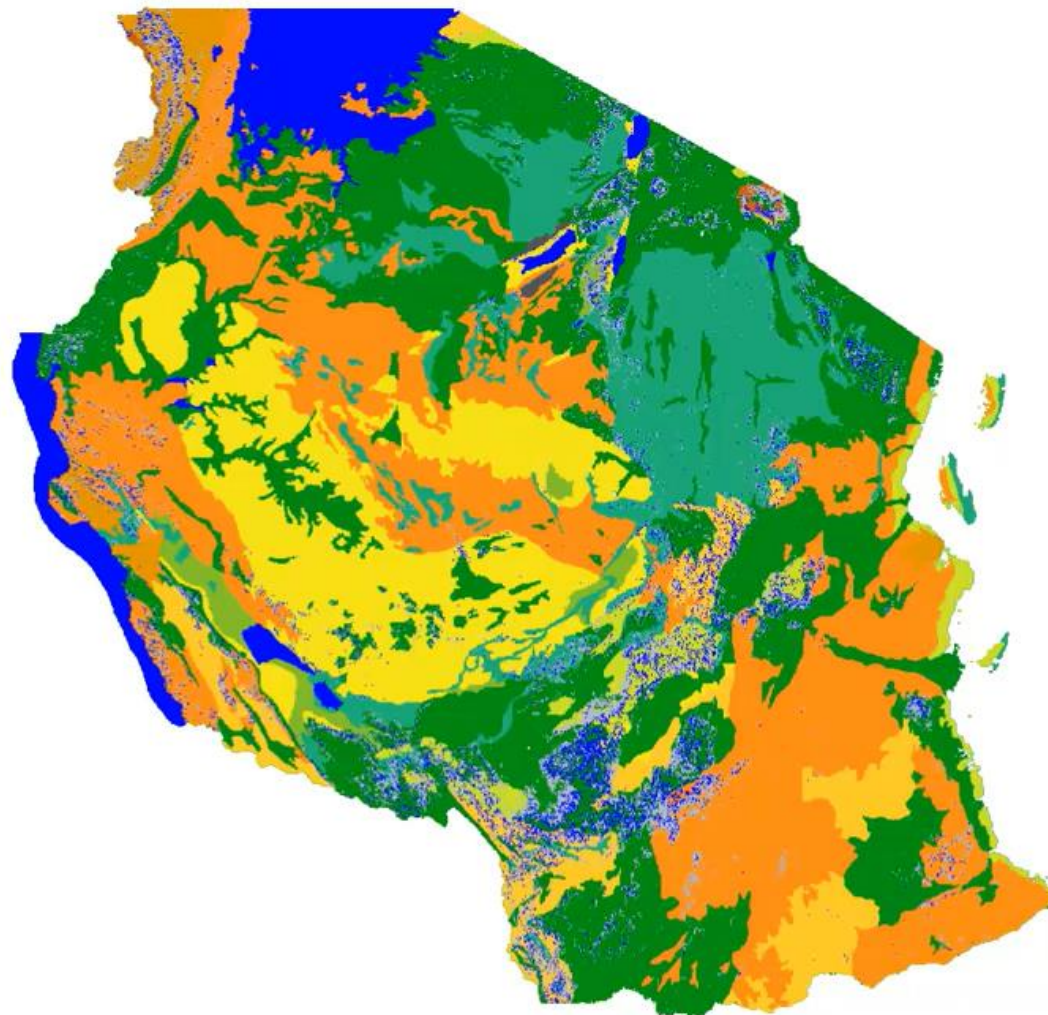


Module 4



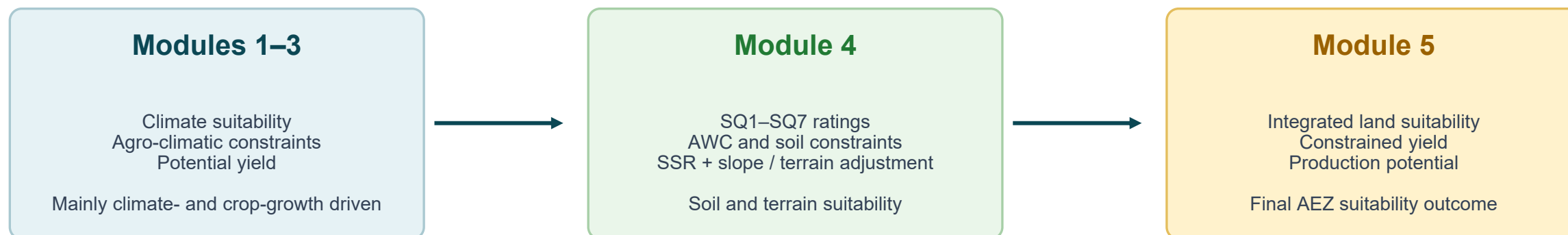
Maize_SQ_SSR_RF_conv_high_tanzania_rasters

Maize_SQ_SSR_RF_conv_high_tanzania_tables



From agro-climatic potential to integrated land suitability

Modules 1–3 estimate the climate-based production potential. Module 4 adds soil and terrain constraints. Module 5 integrates both dimensions for final suitability and yield assessment.



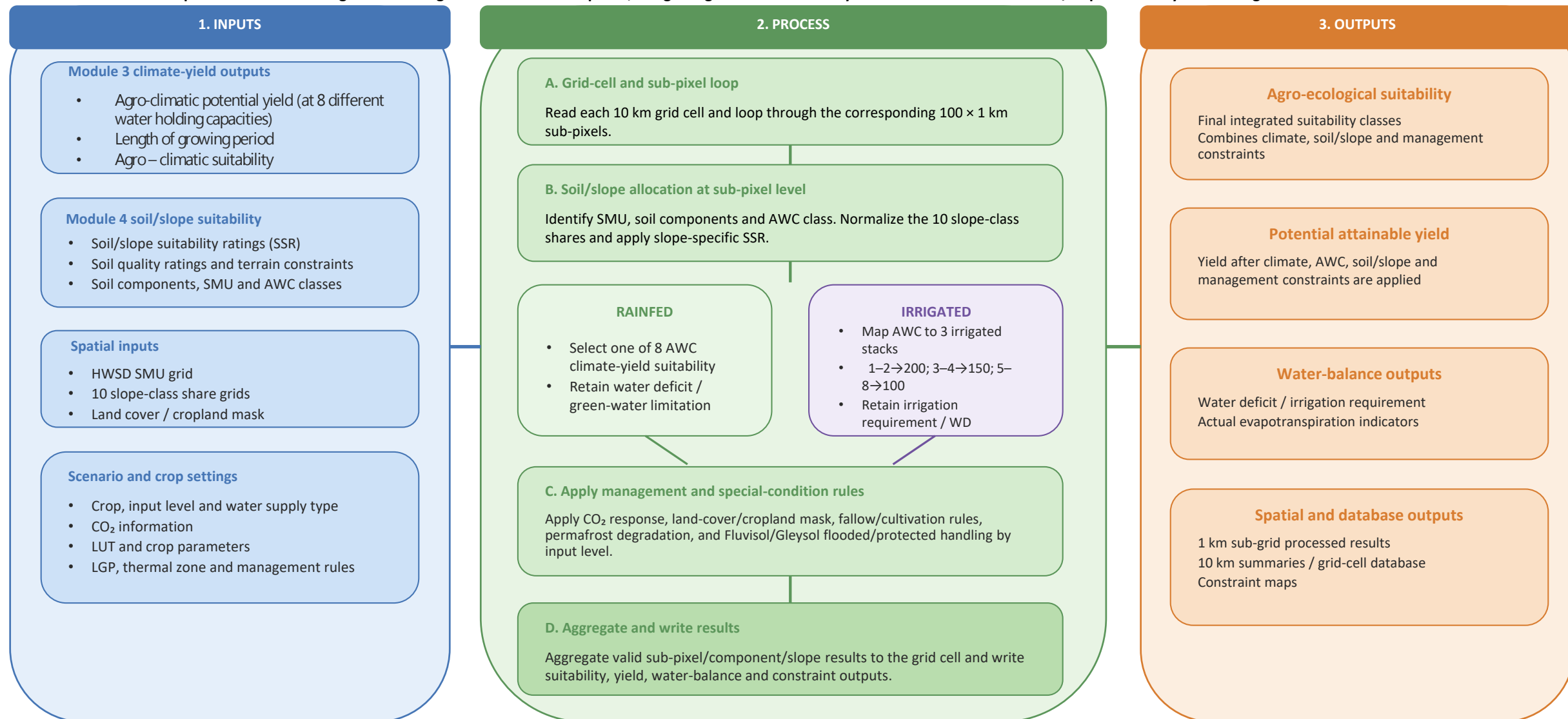
Key information carried forward



Key message: Module 5 adjusts the agro-climatic potential yield using soil suitability and terrain limitations to generate the final AEZ suitability outcome.



Module 5 processes each 10 km grid cell through its 100 × 1 km sub-pixels, integrating Module 3 climate-yield results with Module 4 soil/slope suitability and management conditions.

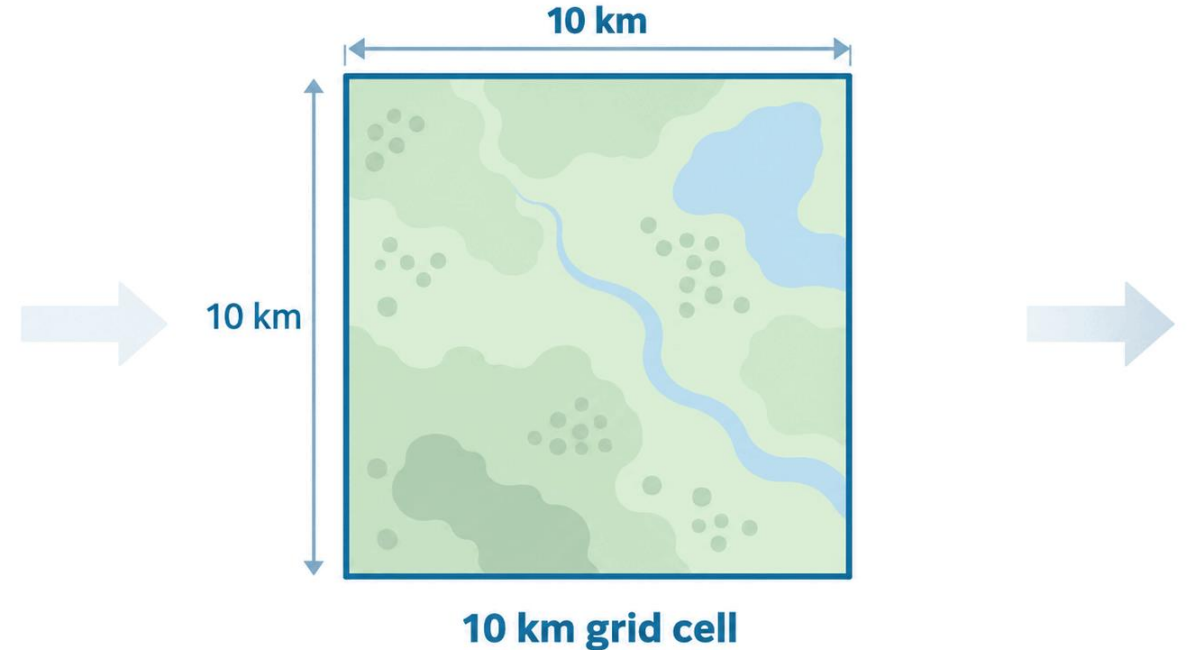


Rainfed and irrigated pathways differ because Module 5 selects different AWC-indexed climate-yield stacks before combining them with local soil/slope suitability and management rules.

Spatial-resolution challenge in Module 5

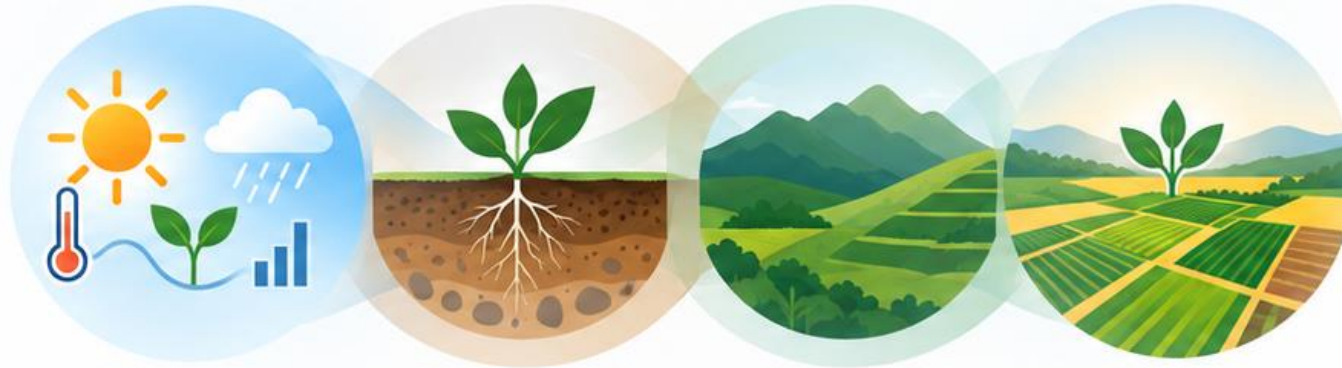
CAVA climate data at 1 km (preferred) or 10 km
25 km → 1 km = awkward assignment to 1 km subpixels
10 km → 1 km = clean nested integration

Current Resolution of CAVA: 25 km



1

Step 1: Start from one 10 km grid cell



Thank You
